#5: Self-Supervised Drone Anomaly Detection via Joint Embedding Predictive Architecture (JEPA)

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Abstract

We propose **JEPA-DRONE**, a self-supervised anomaly detection framework for UAV flight trajectories requiring **zero labeled fault data**. We adapt the Joint Embedding Predictive Architecture (JEPA) [3] to GPS telemetry and introduce *entropy-guided adaptive masking* (20–50% per chunk) as the core novelty. An *Isolation Forest* [4] on the resulting 256-D embeddings detects faults and GPS spoofing at inference with no retraining. Targets: **AUC-ROC** $>$ **0.90**, Recall $>$ 85%, FAR $<$ 5%—approaching a fully supervised upper bound at zero annotation cost.

1 Problem Statement & Motivation

UAVs in safety-critical settings (medical delivery, search-and-rescue, autonomous fleets) face catastrophic mid-flight faults: motor failure, GPS dropout, battery depletion, and adversarial GPS spoofing. The core obstacle is the **labeled-data bottleneck**—real crashes are rare, proprietary, and dangerous to collect, making supervised detectors impractical. Rule-based systems [1] miss complex correlated faults; LSTM autoencoders [2] fall $\sim$0.23 AUC short of supervised bounds. We close this gap with purely self-supervised representation learning.

2 Proposed Approach

Pipeline. Raw flight logs (9 features/waypoint at 4 Hz) are tokenized into 20-waypoint (5-second) chunks. The **entropy-guided adaptive masking** module selects 20–50% of each chunk: low-entropy (stable cruise) chunks receive a harder 50% mask; high-entropy (turbulent) chunks receive a lighter 20% mask. The JEPA context encoder (f_θ, 3-layer MLP, 256-D) predicts latent embeddings of masked waypoints from unmasked context—**no labels required**:

$$\mathcal{L} = \frac{1}{|\mathcal{M}|} \sum_{i \in \mathcal{M}} \left\| g_\phi(\mathbf{z}_{\text{ctx}}, i) - \text{sg}(f_{\bar{\theta}}(\mathbf{C}_k)_i) \right\|_2^2$$

At inference, an *Isolation Forest* fitted on unlabeled normal-flight embeddings scores each chunk; deviations trigger an alert. **GPS spoofing** is detected *zero-shot*—spoofed coordinates violate learned dynamics, producing embedding-space outliers without any retraining or attack labels.

3 Datasets, Baselines & Expected Results

Datasets:
- Kaggle DJI: 1,000+ real flights ($\sim$120K chunks)
- MidAir Simulation: 420K+ frames
- Synthetic faults: 50K chunks, 7 types (motor lockout, GPS dropout, battery failure, bias drift, oscillation, wind, GPS spoofing)

Baselines:
- Random (lower bound)
- Rule-Based Threshold [1]
- LSTM Autoencoder [2]
- Supervised GBT† (upper bound; requires labeled crash data)

Expected results (Table 3). JEPA-DRONE targets $\approx$95% of the supervised bound at zero annotation cost.

Method	AUC	Rec	Prec	FAR	F1
Random	0.50	50%	—	$>$20%	—
Rule-Based	0.65	60%	62%	15%	0.61
LSTM-AE	0.72	65%	68%	18%	0.67
Sup. GBT†	0.95	92%	93%	3%	0.93
JEPA-DRONE $\star$	**$>$.90**	**$>$85%**	**$>$80%**	**$<$5%**	**$>$.87**

Table 3: $\star$=ours; †=requires labeled crash data. Metrics: AUC-ROC, Recall, Precision, FAR, F1.

Wks	Milestone	Lead
1–2	Data pipeline (DJI, MidAir, synthetic)	Yaswanth R.
3–4	JEPA encoder; fixed-mask baseline	Ubaid K.
5–6	Adaptive masking; mask ablation	Siva Sai D.
7–8	*Isolation Forest* + GPS spoofing experiments	Ubaid K.
9–10	Full evaluation & ablations	All
11–12	Report, slides, code release	All

Table 2: Timeline & Team Responsibilities.

References

[1] N. Avdelidis, A. Tsourdos, P. Whidborne. Model-based fault diagnosis for UAV actuators. *IEEE/AIAA DASC*, 2020.

[2] P. Malhotra, L. Vig, G. Shroff, P. Agarwal. LSTM-based encoder-decoder for multi-sensor anomaly detection. *NeurIPS Workshop on Time-Series*, 2022.

[3] M. Assran et al. Self-supervised learning from images with a joint-embedding predictive architecture. *CVPR*, 2023.

[4] F. T. Liu, K. M. Ting, Z.-H. Zhou. Isolation forest. *ICDM*, 2008.